



Andreia Mendes

Nationality: Portuguese **Date of birth:** 16/12/1998 **Place of birth:** Portugal **Gender:** Female

Phone number: (+351) 927416449 **Email address:** andreia.fer.mendes@gmail.com

ABOUT ME

I am a passionate programmer with a strong interest in game development, cultivated during a summer internship where I first encountered the world of coding. Since then, I've been captivated by the challenge of solving complex problems and the endless possibilities programming offers.

I hold a degree in game development, during which I have worked on a variety of projects. Notably, I developed a VR tool to assist in the therapy of schizophrenia, collaborating with a medical team to ensure the game's design was suitable for patients. I also conducted research on XR technologies for a train simulator project, exploring various immersive experiences and their applications in realistic simulations. My experience extends to the development of customizable avatars and a materials blending application, where I created a shader that allows users to define and blend different materials. Additionally, I researched real-time cloth simulation, tackling the challenges of optimizing physics calculations for web environments.

In my free time, I enjoy playing video games, particularly RPGs, puzzles, open-world games, and survival titles. This passion for gaming inspires my work and drives my commitment to honing my skills. I thrive on challenges and find satisfaction in accomplishing tasks I enjoy.

Although English is not my first language, I communicate comfortably with it on a daily basis, engaging with people around the world. I am always eager to learn new techniques and expand my expertise, recognizing the importance of continuous growth in the ever-evolving programming landscape. I also value acquiring skills beyond the gaming scope, as they significantly contribute to my overall personal and professional development.

EDUCATION AND TRAINING

Cambridge certificate CAE level (C1)

Cambridge [2015]

City: Porto | Country: Portugal

Ciências e Tecnologias (Sciences and Technologies)

Didaxis Vale S. Cosme [01/09/2007 – 31/07/2016]

Address: Av. Tibães 1199, Vale de São Cosme, 4770-568 Vila Nova de Famalicão (Portugal) | Website: <https://www.didaxis.pt/regular/>

Licenciatura em Engenharia e Desenvolvimento de Jogos Digitais (Degree in Engineering and Development of Digital Games)

Instituto Politécnico do Cávado e Ave (IPCA) [19/09/2016 – 31/07/2019]

Address: Campus do IPCA - Lugar do Aldão, 4750-810 Vila Frescainha (São Martinho), 4750-810 Barcelos (Portugal) | Website: www.ipca.pt

Mestrado em Engenharia e Desenvolvimento de Jogos Digitais (Masters in Engineering and Development of Digital Games)

Instituto Politécnico do Cávado e Ave (IPCA) [19/09/2019 – 31/08/2022]

Address: Campus do IPCA - Lugar do Aldão, 4750-810 Vila Frescainha (São Martinho), 4750-810 Barcelos (Portugal) | Website: www.ipca.pt

WORK EXPERIENCE

Engineering assistant

Bosch [01/07/2016 – 31/08/2016]

City: Braga | Country: Portugal

First work experience in an industrial environment through a summer internship at Bosch. Assisted with data management in Excel while gaining exposure to the operations of a large company. This experience also marked my first introduction to programming through interactions with the IT department, inspiring me to pursue a degree in the field.

Application programmer

Instituto Politécnico do Cávado e Ave (IPCA) [01/03/2021 – 31/01/2023]

City: Barcelos | Country: Portugal

Work internship facilitated by IPCA in collaboration with a medical team, focused on developing a VR serious game for schizophrenia therapy and/or diagnosis. This project later became the subject of my master's thesis.

Software developer

CCG/ZGDV Intitute [01/10/2023 – Current]

City: Guimarães | Country: Portugal | Website: <https://www.ccg.pt/> | Name of unit or department: Computer Vision, Interaction and Graphics (CVIG) - Business or sector: Professional, scientific and technical activities

Development of AR/VR solutions and conducting research for companies. Finding creative solutions to complex problems and pushing the boundaries of AR/VR technology to drive innovation. Tested applications on devices such as smartphones, Oculus Quest 2 & 3, and HoloLens. Experienced leveraging compute shaders to optimize performance and enhance immersive experiences.

LANGUAGE SKILLS

Mother tongue(s): Portuguese

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1

SPOKEN INTERACTION C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

DIGITAL SKILLS

Programming

C# / C++ / C / SQL / GLSL/HLSL / Swift-Xcode (Basic) / JavaScript

Modelling Software

Maya Software / 3DS Max Software / Blender Software

Software

Unity / Visual Studio / Windows Forms / SQL Server / GitHub Desktop / Microsoft: Microsoft Excel, Microsoft PowerPoint, Microsoft Word, Microsoft Outlook / Monogame framework / AR / AR Core / VR / MR TK / OVR / XR Toolkit

PROJECTS

[2018]

SQL server integration with Windows Forms

Developed a SQL Server database and a complementary Windows Forms application for managing a shop-themed system, including sales data, customer information, and product details. The application featured pre-built queries, such as adding new sales, and allowed users to send custom queries to the connected SQL Server. Query results were displayed in tables, enabling tasks like filtering product sales above a certain price. Built using SQL Server, Visual Studio, and Windows Forms, and programmed in C# and SQL.

[2018]

AR Mobile Game

Completed a course in AR development, where I learned to create AR games using ARCore, a software system that detects surfaces and markers via camera systems. However, due to hardware incompatibility, I explored an alternative approach by leveraging the phone's gyroscope to simulate an AR experience. In my project, players had to physically turn their phones to locate and interact with creatures appearing around them. While the camera was used for cosmetic purposes, the core gameplay relied on gyroscope-based orientation. The game featured increasing difficulty, with enemies becoming more numerous and challenging over time.

This innovative approach, diverging from traditional AR techniques, was praised during academic evaluations for its originality and immersive design. The project was developed using Unity and programmed in C#.

[2019]

Randomly generated terrain using multithreading for a game

For this project, I combined concepts from two university courses—Artificial Intelligence and Optimization Techniques—to develop a terrain generation system using noise algorithms and multithreading. The goal was to efficiently create a unique, playable environment each time it was generated.

Terrain generation was achieved using Perlin noise, manipulating a plane mesh to form dynamic landscapes. A PNG image was also generated, assigning colors based on biome, height, neighboring biomes, and a chaos factor to introduce diversity. To enhance efficiency, I implemented multithreading, reducing generation time from approximately 2 minutes to 30 seconds—making calculations nearly four times faster.

Later, I expanded the project by incorporating concepts from my Game Projects course. The generated terrain and multithreading optimizations became the foundation for a full game, featuring creature and object generation, inventory systems, player movement, and cave generation using a recursive backtracker algorithm for labyrinth creation.

This project was developed in Unity and programmed in C#.

[2020]

Game engine

For this project, I built my own game engine using the MonoGame framework instead of relying on a pre-existing engine like Unity. My goal was to create a highly optimized engine that maximized performance. To evaluate its efficiency, I implemented a test scenario rendering a spaceship model. The objective was to render as many instances as possible while maintaining a stable frame rate. The ships moved in a set direction and were destroyed upon reaching a certain distance.

To optimize performance, I employed techniques such as object pooling, optimized data structures, and rendering optimizations like culling unnecessary triangles. As a result, my modest computer, equipped with a GeForce 965M graphics card, successfully rendered over ten thousand ships while maintaining 60+ FPS for an entire hour.

This project provided deep insight into optimization and demonstrated the power of efficient code. Ironically, the same computer now struggles with certain unoptimized software, highlighting the importance of performance tuning.

Developed using Visual Studio and MonoGame, and programmed in C#.

[01/03/2021 – 31/01/2023]

Developement of a VR game for the therapy and/or diagnosis of Schizophrenia

Schizophrenia is a severe mental disorder that distorts a person's perception of reality, often necessitating lifelong treatment. In recent years, innovative treatment methods, including serious games and VR technologies, have emerged. This project aims to develop a VR tool to assist in the therapy of schizophrenia patients, tested under medical supervision to evaluate its viability as an alternative to traditional therapy due to its ease of use and convenience.

Collaborating with a medical team, we designed game tasks while being mindful of the environments and exercises presented to patients. For example, to avoid triggering paranoia—which is common among schizophrenia patients—we deliberately excluded mirrors from the environment, as patients may misinterpret their reflections as strangers.

Initially, we aimed for the game to be compatible with multiple VR systems. However, this proved challenging due to the lack of universal software across different platforms. As a result, we shifted focus to creating the most immersive experience possible, ultimately choosing the Oculus Quest 2, a wireless system that supports hand tracking and is among the most accessible VR options.

We developed the game using the Unity platform, which is well-suited for Oculus Quest 2 development and offers ready-to-use frameworks. During development, we explored three frameworks: XR Toolkit, OVR Toolkit, and Mixed Reality Toolkit (MRTK). Initially, we used both XR and OVR, with XR supporting broader VR system integration, while OVR was used for hand tracking. Eventually, we transitioned from XR to MRTK for compatibility with HTC Vive, as XR was not suitable.

Ultimately, we decided to focus solely on the Oculus Quest 2 and hand tracking to ensure an optimal user experience. I was responsible for the majority of the game's development, including UI, game flow, mechanics, and VR integration. This project was a significant challenge for me, as I had no prior experience in VR development, but I embraced the opportunity to learn and grow.

Link: [https://www.researchgate.net/publication/](https://www.researchgate.net/publication/366353663)

[366353663 Sensors Transducers Development of a Virtual Reality Tool for Therapy and Diagnosis of Schizophrenia](https://www.researchgate.net/publication/366353663)

[2024 – Current]

Developement of a realistic and customizable human avatar

Developed realistic human avatars that users can customize through sliders for various physiological aspects, such as arm and leg length, height, body weight, and more. Each aspect is represented internally as model information, defining the minimum and maximum values for each slider, which are then blended based on user input.

The most challenging aspect of this project was optimization, as it needed to run efficiently in a web environment. The initial algorithm iteration was executed on the CPU, but modern techniques shifted the processing to the GPU to utilize its idle capacity and free up the CPU for other calculations.

By implementing compute shaders and multithreading, we significantly accelerated the blending process from just under 20 seconds to a fraction of a second. This optimization enabled real-time customization of avatars without lag.

The customizable avatar component is part of a larger prototype and has been successfully integrated into the overall project.

[2024 – Current]

Prediction of thread/sample aspect

Developed an application capable of estimating the final appearance of blended materials. For this project, I created a customizable shader that allows users to define and create the materials they want to blend. I also developed an algorithm that blends these materials based on varying weights to produce a final product visualization. Key aspects such as smoothness, reflection, and color were carefully considered to ensure accurate representation of the blended materials.

[2024 – Current]

Development of a train simulator in XR

Conducted research and development of a prototype for a realistic train simulator, featuring real-world tracks and trains. The simulator supports various levels of immersion, from traditional on-screen simulation to full surround and VR experiences, with control options ranging from virtual interfaces to the recreation of actual train controls.

Research indicated that using OpenRails as a foundation is viable due to its open-source nature, which already includes physics for different types of trains, train systems, and track systems, as well as tools for track building. Additionally, I developed realistic shaders to simulate weather effects, such as fog, rain, and snow.

[2024 – Current]

Cloth and clothes simulation

Conducted research and development of a tool capable of simulating cloth and clothing realistically in real time on a human model. Realistic cloth simulation is challenging due to the complex nature of fabric behavior, which involves factors such as elasticity, friction, and interaction with environmental forces like wind and gravity. Additionally, simulating the deformation of these complex structures requires precise

calculations of how the fabric moves and responds to the underlying human model, which adds significant computational load.

The most challenging aspect of this project is optimization, as physics simulation is resource-intensive, and realistic cloth physics involves simulating complex deformable structures. Furthermore, since this tool is intended to run in a web environment, available computational resources are further limited.

Various methods are being evaluated for this project, including the use of compute shaders and different physics simulation techniques, such as spring-mass and constraint-based simulations.